SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14 COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: *Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities.* *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:100

1.A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.

2.The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.

3.A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

4.An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)**and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in
the web application environment and recommend measures to minimize exposure

5.The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Code of Conduct:

I R Dinesh-192421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dinesh R

1. Enumeration of scanme.nmap.org

Aim

To identify the active host, open TCP ports, running services, service versions, and operating system information using Nmap.

Commands

nmap -sn scanme.nmap.org
nmap -sS -sV -O scanme.nmap.org
nmap -sC -sV -O scanme.nmap.org

Analysis

1. **Host Discovery:** Determines whether the target host is reachable.
2. **TCP Port Scanning:** Identifies open, closed, and filtered ports.
3. **Service Version Detection:** Identifies the services and their versions.
4. **OS Fingerprinting:** Estimates the operating system based on network responses.
5. **Default Script Scanning:** Checks common service configurations and identifies possible security issues.

Risk Assessment

Finding	Business Purpose	Risk
HTTP (80)	Hosts web pages	Medium
HTTPS (443)	Provides secure web access	Medium
SSH (22)	Remote administration	Medium
Other open ports	Depends on the service	Requires investigation

These are examples of common services, not verified findings for the target.

Attack Surface Analysis

Attackers may use enumeration results to identify outdated software, weak authentication, and vulnerable services. This information could help them select targets for further vulnerability analysis or brute-force attempts.

Recommendations

- Disable unnecessary services and ports.
- Update the operating system and applications.
- Use strong passwords and SSH key authentication.
- Apply firewall rules and monitor network traffic.
- Perform regular security assessments.

Result

Nmap can identify exposed services and help administrators assess the server's attack surface.

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2. Enumeration of Metasploitable 2 (192.168.56.101)

Aim

To identify open ports, running services, service versions, and potential security weaknesses on the intentionally vulnerable Linux server.

Commands

nmap -sn 192.168.56.101
nmap -sS -sV -O 192.168.56.101
nmap -sC -sV -O 192.168.56.101

For a comprehensive TCP port scan in the isolated lab:
nmap -p- -sV 192.168.56.101

Analysis

Metasploitable 2 is intentionally vulnerable and commonly includes services such as FTP, SSH, Telnet, SMTP, HTTP, SMB, and database services.

Service	Port	Purpose	Risk
FTP	21	File transfer	High
SSH	22	Remote access	Medium
Telnet	23	Remote administration	High
SMTP	25	Email transfer	Medium
HTTP	80	Web hosting	High
SMB	139/445	File sharing	High
MySQL	3306	Database access	High

These are typical examples; confirm the actual open ports and versions using Nmap.

Risk Assessment

- **High Risk:** Exposed vulnerable services, insecure file transfer, and poorly secured file-sharing services.
- **Medium Risk:** Services that require authentication but may have weak configurations.
- **Low Risk:** Necessary services with restricted access and secure configurations.

Attack Surface Analysis

A penetration tester can use the identified service versions to research known vulnerabilities and determine which services require further authorized testing.

Recommendations

- Disable unnecessary services.
- Replace Telnet and FTP with secure alternatives.
- Update vulnerable applications.
- Restrict access using firewall rules.
- Use strong authentication and monitor logs.

Result

Enumeration helps identify vulnerable services and prioritize security testing.

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3. Enumeration of Windows Workstation (192.168.56.102)

Aim

To identify active ports, Windows services, file-sharing protocols, remote management services, and potential security risks.

Commands

nmap -sn 192.168.56.102
nmap -sS -sV -O 192.168.56.102
nmap -p 135,139,445,3389,5985,5986 -sV 192.168.56.102

Analysis

Port	Service	Purpose	Potential Risk
135	MS RPC	Windows communication	Information disclosure
139	NetBIOS	Network communication	Network information exposure
445	SMB	File and printer sharing	Unauthorized access
3389	RDP	Remote desktop	Brute-force attacks
5985	WinRM HTTP	Remote management	Unauthorized remote access
5986	WinRM HTTPS	Secure remote management	Misconfiguration or weak access control

The table lists services to check, not confirmed scan results.

Least Privilege Analysis

Only services required for the workstation's business purpose should be enabled. File sharing and remote management should be restricted to authorized users and networks.

Security Risks

- SMB misconfiguration may expose shared files.
- Weak RDP passwords may allow account compromise.
- Exposed remote management services may enable unauthorized access.
- Attackers who compromise one workstation may attempt lateral movement to other systems.

Recommendations

1. Disable unnecessary network services.
2. Restrict SMB and RDP using firewall rules.
3. Enable Network Level Authentication for RDP.
4. Apply Windows security updates.
5. Use strong passwords and multifactor authentication where supported.
6. Monitor authentication logs and network activity.

Result

Nmap enumeration helps determine whether the Windows workstation exposes unnecessary services before it connects to the production network.

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4. Enumeration of OWASP Juice Shop (192.168.56.103)

Aim

To identify the network services and web interfaces exposed by the OWASP Juice Shop server.

Commands

nmap -sn 192.168.56.103
nmap -sS -sV -O 192.168.56.103
nmap -sC -sV -p- 192.168.56.103
To identify common web services:
nmap -p 80,443,3000,8080 -sV 192.168.56.103

Analysis

Service	Common Port	Purpose	Potential Risk
HTTP	80	Web application access	Unencrypted communication
HTTPS	443	Secure web access	TLS misconfiguration
Web application	3000	Common Juice Shop deployment port	Application vulnerabilities
Alternate HTTP	8080	Web application or proxy	Unnecessary exposure

Actual ports depend on the server configuration and must be verified.

Attack Surface Analysis

Enumeration reveals exposed web ports, application services, and potentially accessible administrative interfaces. Security testers can use this information to determine which components need further testing.

Potential issues include:

- Outdated web server components.
- Weak administrative access controls.
- Insecure HTTP communication.
- Application vulnerabilities such as injection and broken access control.

Recommendations

1. Use HTTPS with properly configured TLS.
2. Restrict administrative interfaces.
3. Update application dependencies.
4. Apply secure authentication and access controls.
5. Use a firewall to limit exposed ports.
6. Test the application using authorized security testing procedures.

Result

Nmap helps identify exposed web services and supports further security assessment of the Juice Shop environment.

5. Enumeration of Network Gateway (192.168.56.1)

Aim

To identify exposed management interfaces, network services, and potential security weaknesses on the laboratory gateway.

Commands

nmap -sn 192.168.56.1
nmap -sS -sV -O 192.168.56.1
nmap -sC -sV 192.168.56.1

To check common management ports:

nmap -p 22,23,80,443,161 192.168.56.1

Analysis

Port	Service	Purpose	Potential Risk
22	SSH	Secure remote administration	Weak credentials
23	Telnet	Remote administration	Unencrypted communication
80	HTTP	Web management	Credentials may be exposed
443	HTTPS	Secure web management	TLS or access-control weaknesses
161	SNMP	Network monitoring	Information disclosure

These are common ports to investigate, not confirmed findings.

Security Implications

- **SSH:** Weak passwords or excessive access permissions may permit unauthorized administration.
- **Telnet:** Transmits data without encryption.
- **HTTP:** May expose management traffic.
- **HTTPS:** Can still be vulnerable if incorrectly configured.
- **SNMP:** Weak community strings or excessive permissions may expose device information.

Attack Surface Analysis

Attackers may use exposed management interfaces to discover device configurations, identify network information, or attempt unauthorized administrative access. A compromised gateway could put connected systems at risk.

Recommendations

1. Disable Telnet and unnecessary management services.
2. Use SSH with strong authentication.
3. Enable HTTPS for web administration.
4. Restrict management access to trusted administrator IP addresses.
5. Configure SNMP securely and use SNMPv3 where supported.
6. Update gateway firmware regularly.
7. Monitor logs and review firewall rules.

Result

Nmap enumeration helps identify exposed gateway services and supports security hardening to reduce the risk of network compromise.

RUBRICS FOR CALCULATION

Criteria	Weightag (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	P logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	